DEEP ANALYSIS REPORT: DELIVERY TIME PREDICTION USING AI & ML

1. Introduction

This report presents a detailed analysis of a machine learning model for predicting food delivery times. The study includes data preprocessing, visualizations, feature engineering, model evaluation, and operational insights. The goal is to optimize delivery operations by providing accurate time estimates and suggesting data-driven improvements.

2. Data Preprocessing & Feature Engineering

2.1. Data Cleaning & Preparation

- There were no missing values in the datasheet. But certain columns with negative values are cleaned up by removing only that particular row.
- created_at and actual_delivery_time were converted to datetime format.
- New feature: delivery_timetaken_InMinutes = actual_delivery_time - created_at.
- Categorical variables (market_id, order_protocol) were converted to dummies column types and store_primary_category value is replaced with the mean of target variable as the number of unique value is approx. 73 and if I create these number of column then it will create overfitting.
- total_onshift_dashers, total_busy_dashers and total_outstanding_orders are converted to whole numbers.
- OrderCreated_hour and OrderCreated_day_of_week extracted from created_at
- Created a categorical feature 'isWeekend' from OrderCreated_day_of_week.
- Dropped unnecessary columns - created_at, actual_delivery_time.

2.2. Feature Scaling & Transformation

- MinMax Scaling was applied to numerical features for improved model performance.
- Outliers were detected and handled using boxplot analysis.

2.3. Creating training and validation sets -> Used 70:30 ration for train test data split

```
# split data into training and testing sets

# splitting data (70 % training and 30% testing)

x_train, x_test, y_train, y_test = train_test_split(
        x, y, test_size=0.3, random_state=42
)

print(x_train.shape)
print()
print(y_train.shape)
print()
print(x_test.shape)
print()
print(y_test.shape)
print()
```

```
(123043, 15)

(123043,)

(52734, 15)

(52734,)
```

3. Data Visualization & Insights

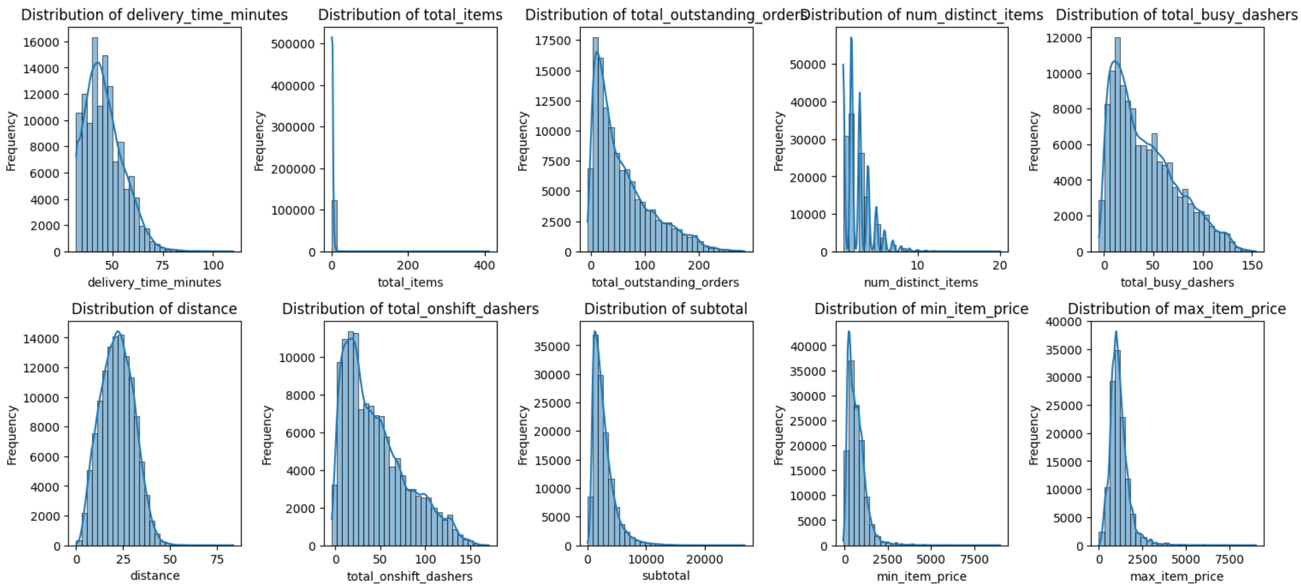
3.1 Features distribution: distribute feature based on numerical and categorical

3.1 Feature Distributions [7 marks]

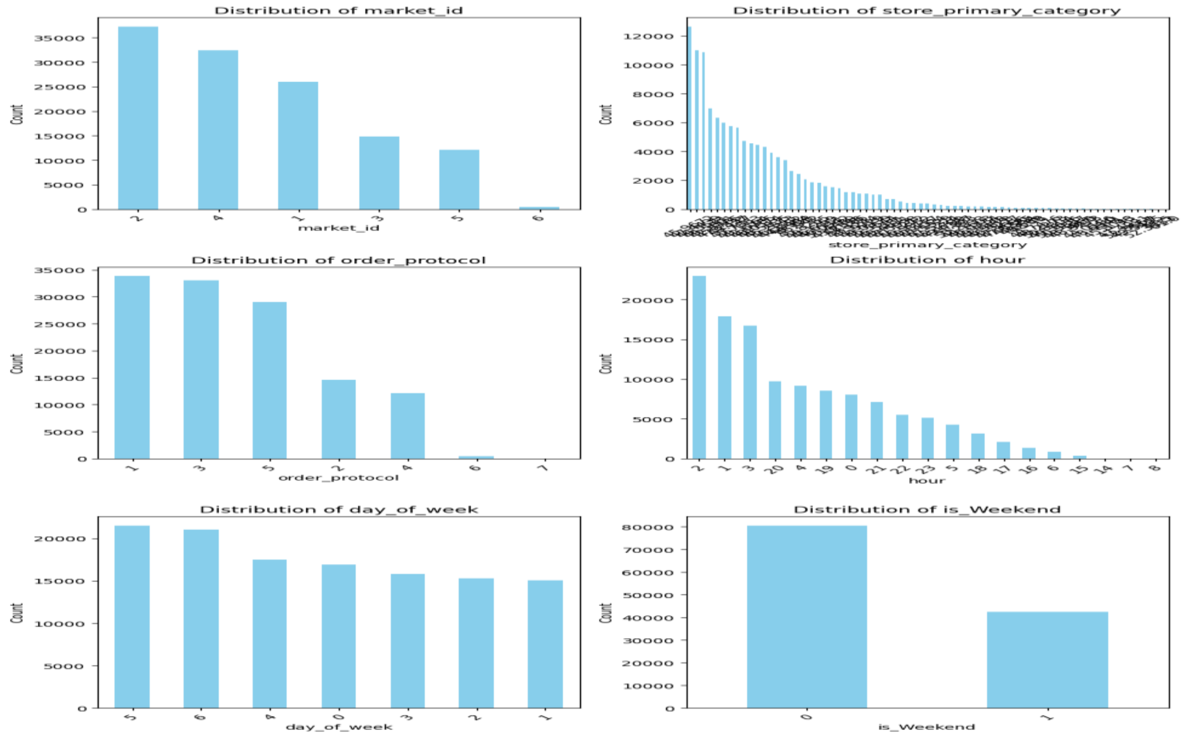
```
[20]: # Define numerical and categorical columns for easy EDA and data manipulation

numerical_cols = ['delivery_time_minutes','total_items','total_outstanding_orders','num_distinct_items',
                  'total_busy_dashers','distance','total_onshift_dashers','subtotal','min_item_price','max_item_price']
categorical_cols = ['market_id','store_primary_category','order_protocol','hour','day_of_week','is_Weekend']
```

3.1.1 Plot distributions for all numerical columns

Histogram of Delivery Time: Shows a right-skewed distribution, indicating a few extreme cases of high delivery time.



3.1.2 Plot Distribution of categorical columns : -



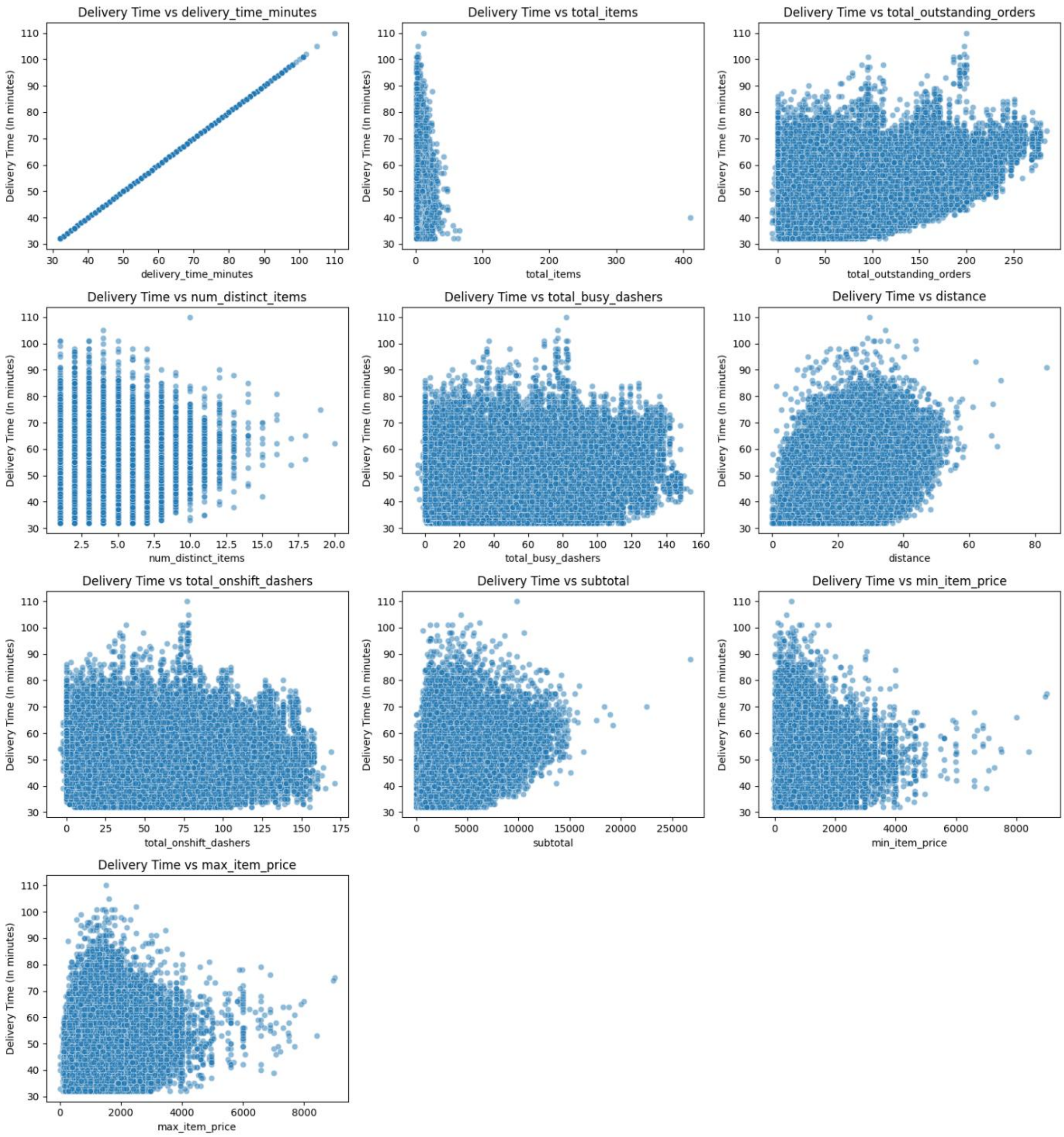
The dataset shows a skewed distribution for market_id with a dominant value of 2, and store_primary_category has a long tail, indicating many categories with few occurrences. order_protocol is heavily concentrated on values 1 and 2, while hour shows peak activity around hours 2 and 3, declining sharply afterward. day_of_week has a relatively even distribution, and is_Weekend shows significantly more occurrences for non-weekend days (0).

3.1.3 Visualise the distribution of the target variable to understand its spread and any skewness:

The histogram displays the distribution of delivery times in minutes, which appears to be right-skewed, indicating that most deliveries take between 35 and 55 minutes. There's a peak frequency around 42-45 minutes, and the curve overlaid suggests an approximate normal distribution centred around this peak, although the actual data deviates with a longer tail towards higher delivery times. This implies that while most deliveries are relatively quick, there are some instances with considerably longer delivery durations

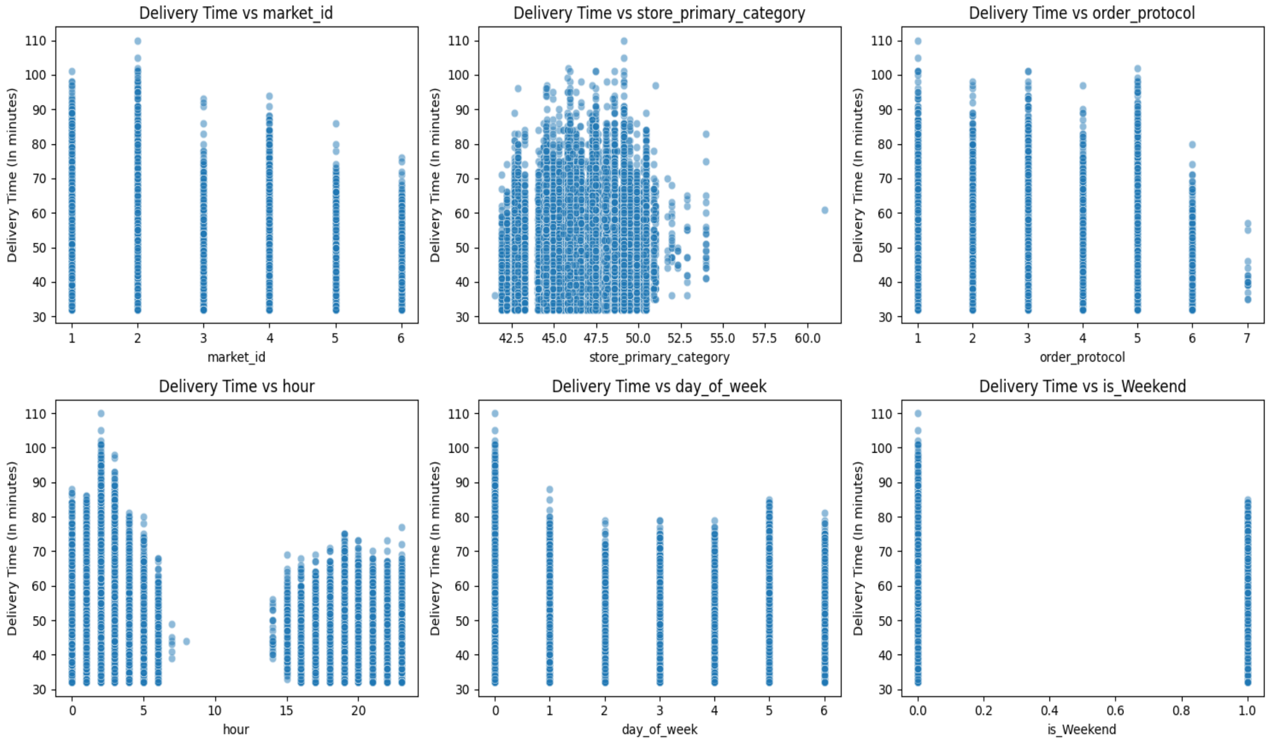
3.2 Relationships Between Features

3.2.1 Scatter plots for important numerical and categorical features to observe how they relate to time_taken



The image displays scatter plots showing the relationship between delivery time (in minutes) and various features. Most plots show a weak or no clear linear correlation, suggesting delivery time is not strongly predicted by individual features like total items, outstanding orders, or number of distinct items. However, some plots, such as those against the number of busy dashers, distance, and subtotals, show a slightly wider spread of delivery times at higher values, hinting at potential indirect influences.

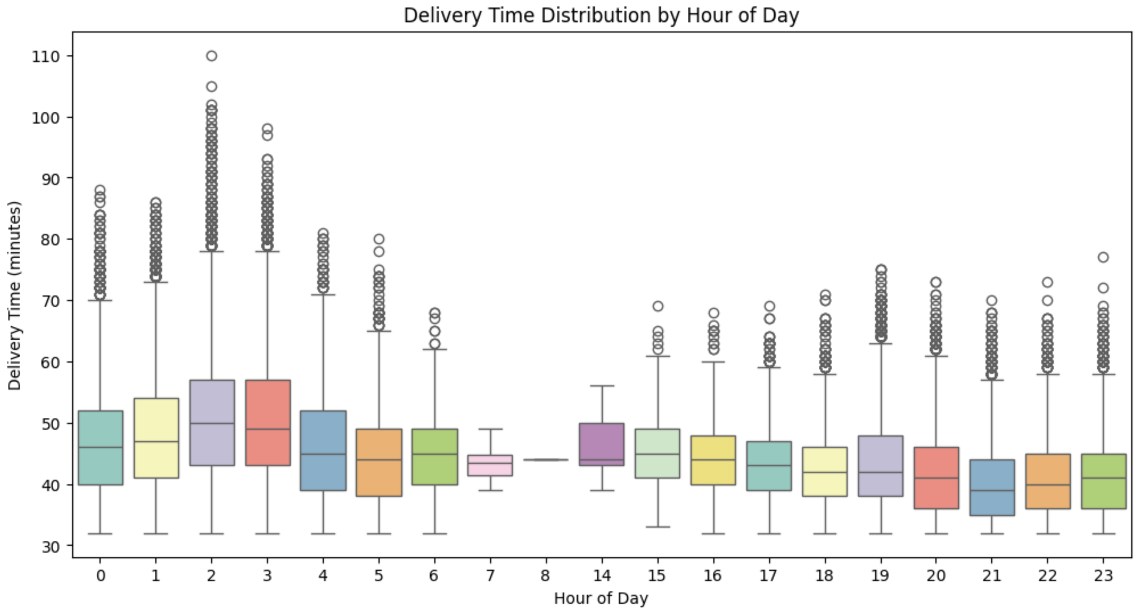
Plot all categorical features against delivery_time_minutes :



The image displays scatter plots examining the relationship between delivery time (in minutes) and categorical features. Delivery time appears to vary across different market_id values, with some markets potentially experiencing longer or shorter average delivery times. Similarly, store_primary_category shows a range of delivery times within each category, with a dense cluster around certain categories.

order_protocol also exhibits variations in delivery time depending on the protocol used. The hour of the day clearly influences delivery time, with longer times potentially occurring during peak hours. day_of_week shows some fluctuation in delivery time across different days, and is_Weekend suggests a possible difference in delivery time between weekends and weekdays.

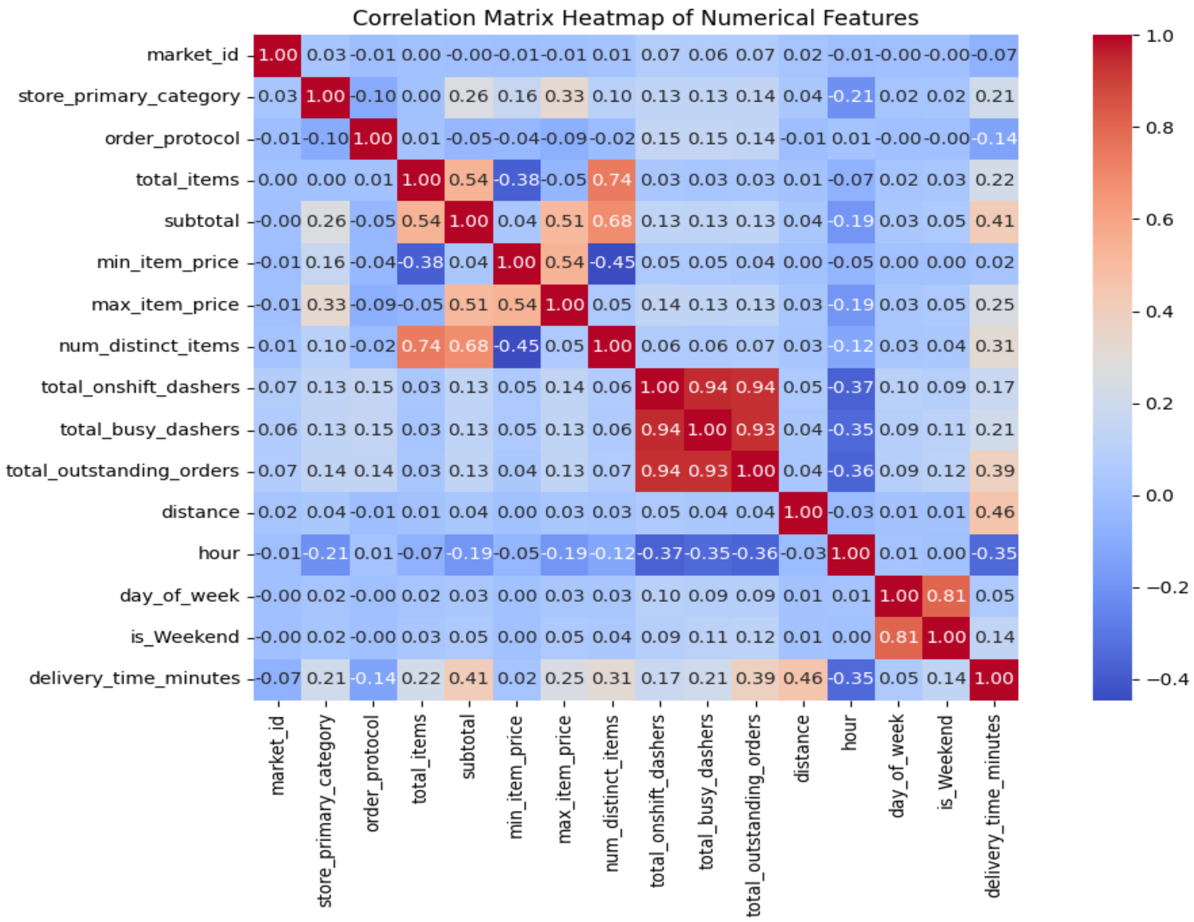
distribution of time taken for different hours :



The box plot illustrates the distribution of delivery times across different hours of the day. Delivery times appear generally higher during the early morning hours (around 0-5) and then decrease significantly during the daytime. There's a noticeable increase in median delivery time again during the evening and late-night hours (around 18-23). The plot also reveals the presence of outliers, especially during the hours with higher median delivery times.

3.3 Correlation Analysis: check correlation between numerical feature with time taken

```
Correlation with delivery_minutes:
market_id                  -0.074076
store_primary_category      0.206624
order_protocol             -0.138156
total_items                 0.217966
subtotal                    0.412902
min_item_price              0.019842
max_item_price              0.253317
num_distinct_items          0.313315
total_onshift_dashers       0.171863
total_busy_dashers          0.206014
total_outstanding_orders    0.385424
distance                    0.460033
hour                       -0.345744
day_of_week                 0.045481
is_Weekend                  0.137693
delivery_time_minutes       1.000000
Name: delivery_time_minutes, dtype: float64
```



The heatmap displays the correlation coefficients between numerical features. Strong positive correlations (red) exist between total_onshift_dashers, total_busy_dashers, and total_outstanding_orders. max_item_price and subtotal also show a positive correlation. Conversely, delivery_time_minutes has a moderate positive correlation with distance and a weak negative correlation with min_item_price. Most other feature pairs exhibit weak or negligible linear relationships.

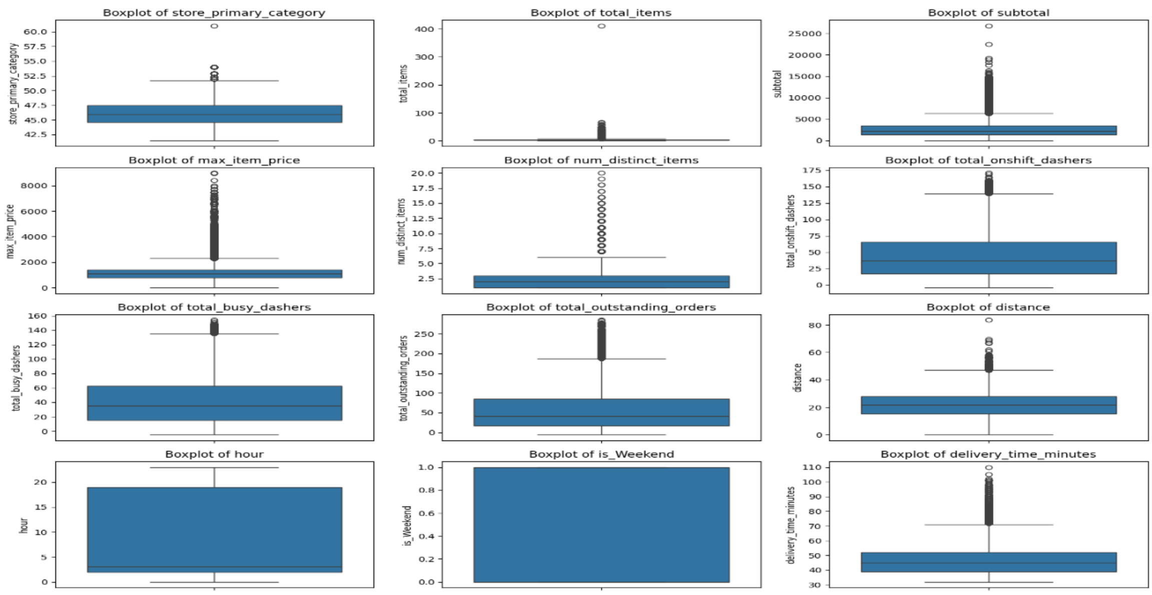
3.3.2 dropped column with weak correlations with the target variable

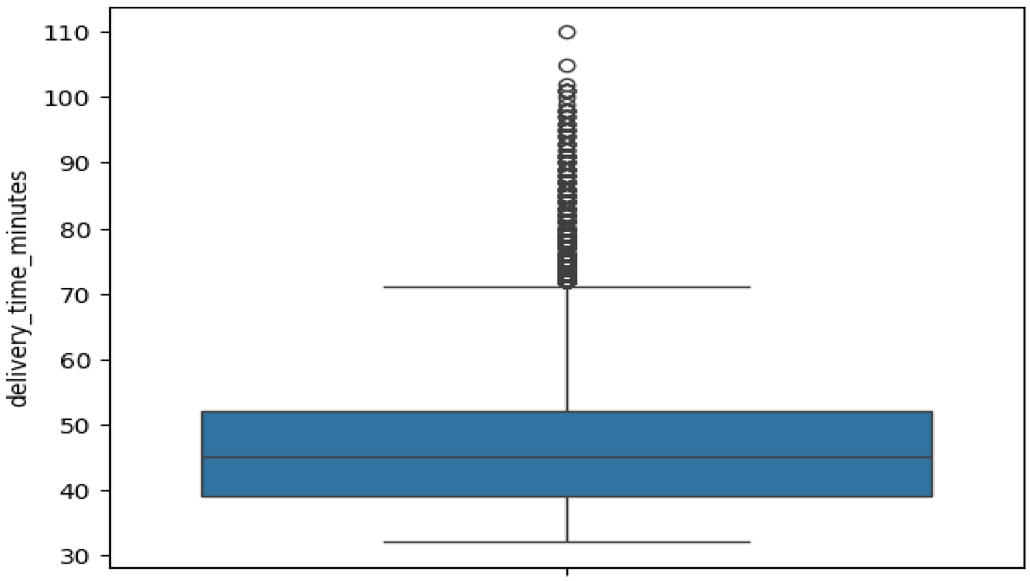
```python
# Drop 3-5 weakly correlated columns from training dataset

weak_features = ['day_of_week', 'min_item_price', 'market_id','order_protocol']

print("Weakly correlated features to drop:", weak_features)


x_train = x_train.drop(columns=weak_features)

# Checking new shape
print("Updated X shape after dropping weak features:", x_train.shape)


x_test = x_test.drop(columns=weak_features)

porter = porter.drop(columns=weak_features)
```

```
Weakly correlated features to drop: ['day_of_week', 'min_item_price', 'market_id', 'order_protocol']
Updated X shape after dropping weak features: (123043, 12)
```

3.4 handling the outliers

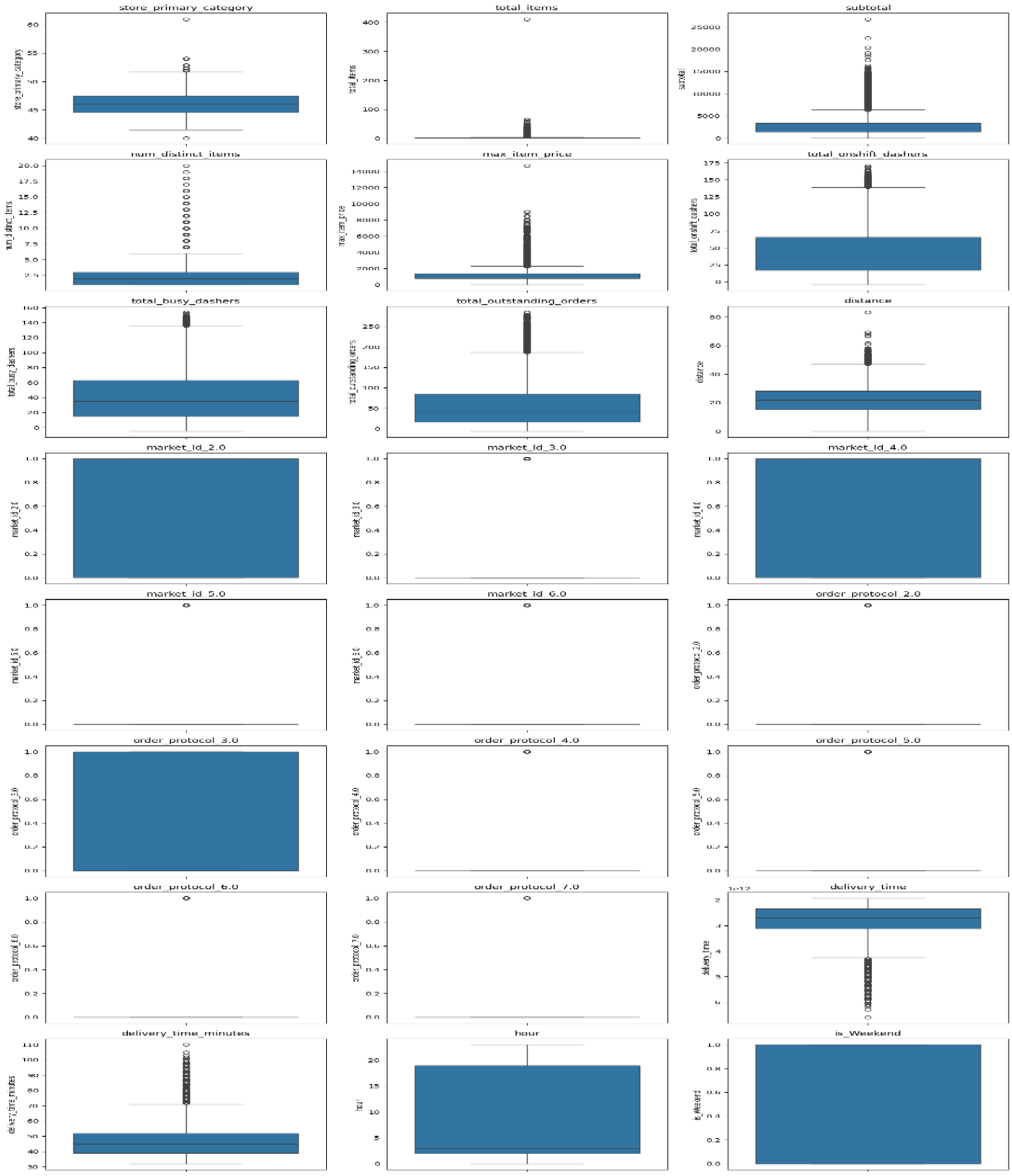
3.4.1 Boxplot for time_taken :





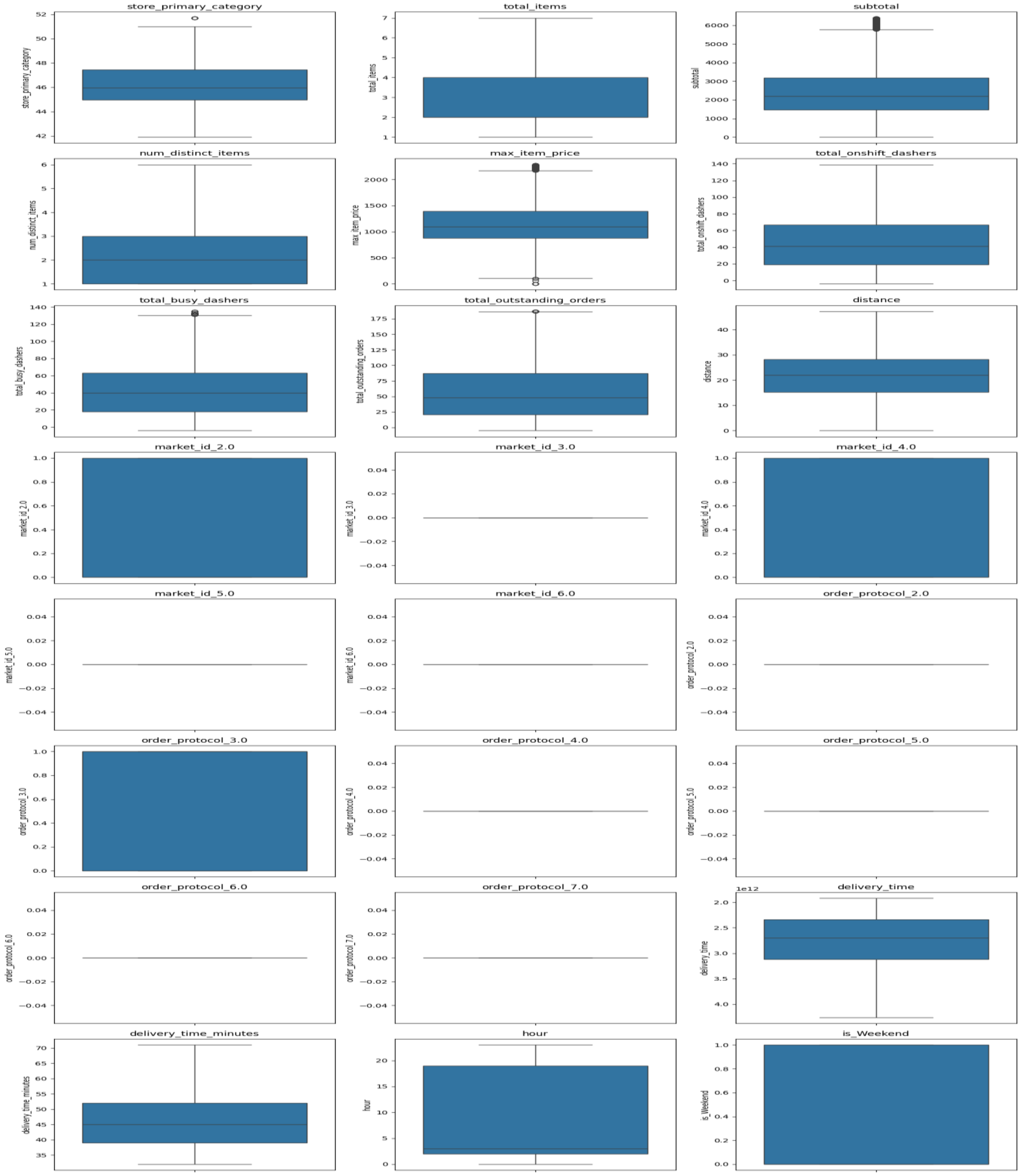
The box plot visualizes the distribution of delivery times in minutes. The central box represents the interquartile range (IQR), with the median delivery time shown by the line inside. The whiskers extend to 1.5 times the IQR, indicating the typical range of delivery times. Numerous outliers are present above the upper whisker, suggesting some deliveries take significantly longer than the majority.

Before removing outliers of all column ->



The image displays multiple box plots showing the distribution of various numerical features. Most features, such as total_items, subtotal, max_item_price, total_onshift_dashers, total_busy_dashers, total_outstanding_orders, and distance, exhibit right-skewness and numerous outliers on the higher end. Features related to market ID and order protocol (when treated as numerical) show a more concentrated distribution around specific values. The target variable, delivery_time_minutes, also shows a right-skewed distribution with several longer delivery times as outliers.

After removing outliers ->



4. Model Training & Performance Evaluation

4.1 Model Selection

Linear Regression was chosen for its interpretability and effectiveness.

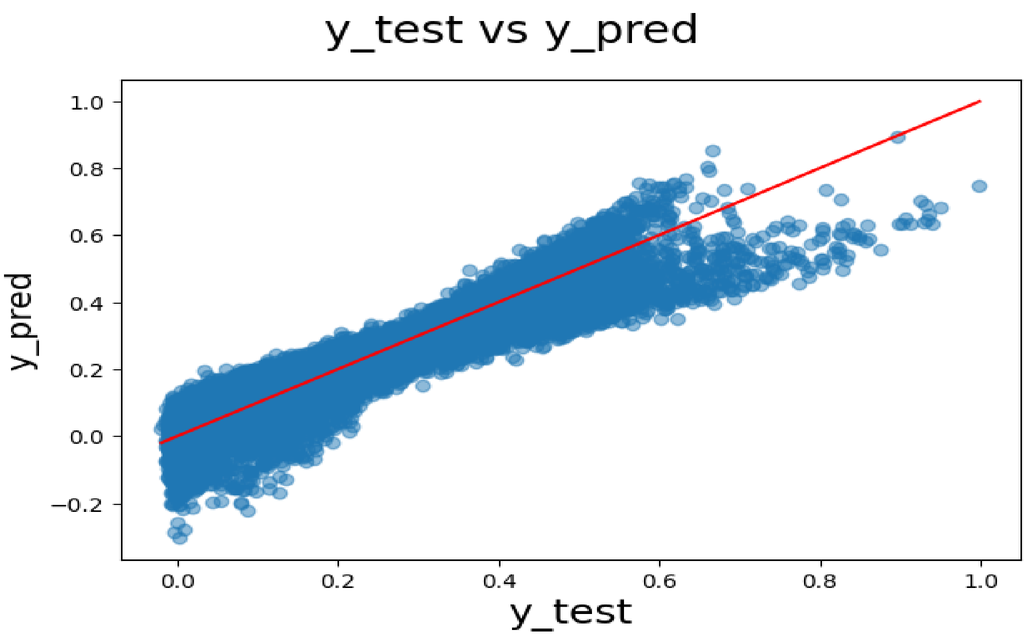
The dataset was split into 70% training and 30% testing sets.

4.2 Final Model - Performance Metrics (Using feature Additions)

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     delivery_time_minutes   R-squared:                       0.903
Model:                               OLS   Adj. R-squared:                  0.903
Method:                    Least Squares   F-statistic:                 2.233e+04
Date:                   Wed, 28 May 2025   Prob (F-statistic):               0.00
Time:                           16:54:19   Log-Likelihood:                 93644.
No. Observations:                  52734   AIC:                        -1.872e+05
Df Residuals:                      52711   BIC:                        -1.870e+05
Df Model:                             22
Covariance Type:               nonrobust
=========================================================================================

===========================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------
const                       0.0474      0.005      9.908      0.000       0.038       0.057
store_primary_category      0.0007      0.000      6.413      0.000       0.000       0.001
total_items                -0.0238      0.007     -3.465      0.001      -0.037      -0.010
subtotal                    0.3571      0.004     93.005      0.000       0.350       0.365
max_item_price              0.1282      0.007     19.299      0.000       0.115       0.141
num_distinct_items          0.1222      0.004     33.482      0.000       0.115       0.129
total_onshift_dashers      -0.8188      0.003   -267.394      0.000      -0.825      -0.813
total_busy_dashers         -0.2965      0.003   -101.420      0.000      -0.302      -0.291
total_outstanding_orders    1.3607      0.003    449.166      0.000       1.355       1.367
distance                    0.3953      0.001    334.596      0.000       0.393       0.398
hour                       -0.0031   2.36e-05   -132.842      0.000      -0.003      -0.003
is_Weekend                  0.0163      0.000     42.791      0.000       0.016       0.017
market_id_2.0              -0.0645      0.001   -111.138      0.000      -0.066      -0.063
market_id_3.0              -0.0599      0.001    -92.088      0.000      -0.061      -0.059
market_id_4.0              -0.0533      0.001    -89.827      0.000      -0.054      -0.052
market_id_5.0              -0.0499      0.001    -71.117      0.000      -0.051      -0.048
market_id_6.0              -0.0388      0.003    -13.394      0.000      -0.044      -0.033
order_protocol_2.0         -0.0101      0.001    -15.924      0.000      -0.011      -0.009
order_protocol_3.0         -0.0196      0.001    -38.823      0.000      -0.021      -0.019
order_protocol_4.0         -0.0239      0.001    -33.880      0.000      -0.025      -0.023
order_protocol_5.0         -0.0379      0.001    -72.924      0.000      -0.039      -0.037
order_protocol_6.0         -0.0226      0.003     -8.006      0.000      -0.028      -0.017
order_protocol_7.0         -0.0239      0.024     -1.011      0.312      -0.070       0.022
==============================================================================
Omnibus:                    19413.693   Durbin-Watson:                   1.999
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           124640.471
Skew:                           1.632   Prob(JB):                         0.00
Kurtosis:                       9.788   Cond. No.                     6.24e+03
==============================================================================
```

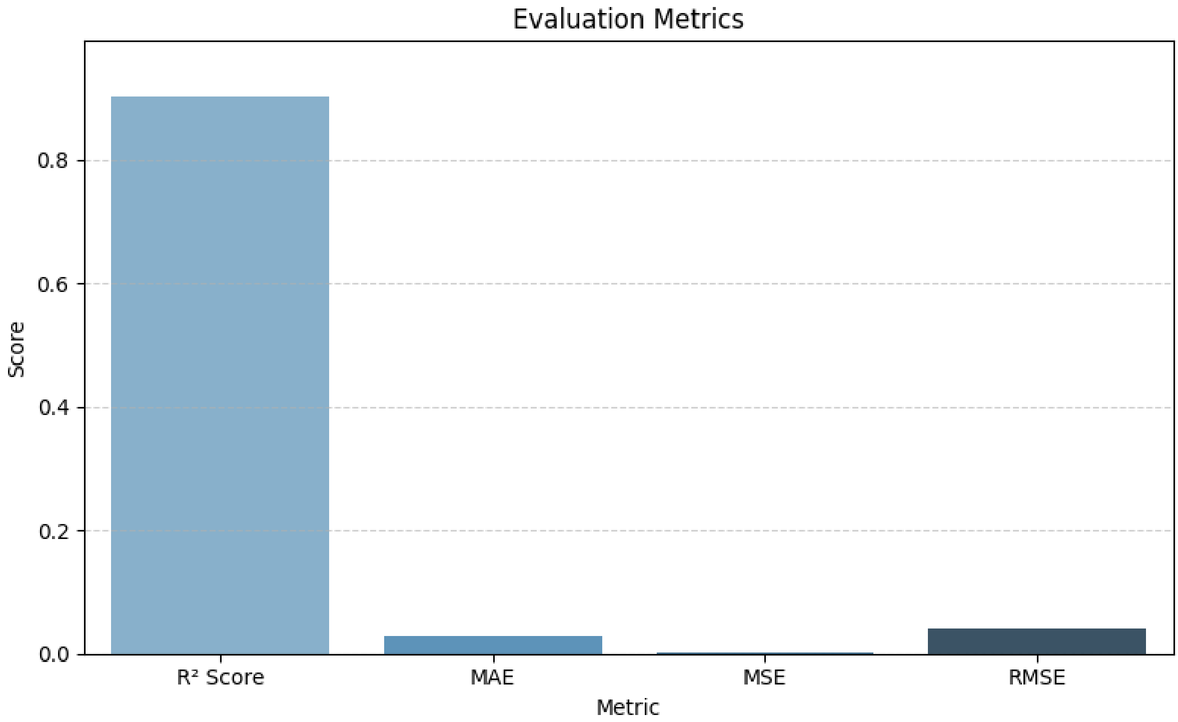
- Model Performance Metrics
  - R-squared (0.903):
    - Indicates that 90.3% of the variance in delivery time is explained by the model's features.
  - Adjusted R-squared (0.903):
    - Close to the R-squared value, confirming that the model is not overfitting despite multiple predictors.

Predict y_test and y_train data :->



The above diagram shows that the scatter plot compares the true values (y_test) against the predicted values (y_pred). The points generally follow an upward trend, indicating a positive relationship between the actual and predicted values. However, there is a noticeable spread around the red diagonal line, which represents perfect predictions. This spread suggests that the model's predictions have some degree of error, with underestimations at lower true values and overestimations at higher true values.

Result of evaluation matrix ->



This bar chart titled "Evaluation Metrics" displays four different metrics: R² Score, MAE, MSE, and RMSE. The R² Score is significantly higher than the other metrics, close to 0.9. MAE and RMSE have very small, positive values, with RMSE being slightly larger than MAE. MSE is almost negligible, appearing close to zero on the graph

4.3 Using RFE: (final model looks like below)

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     delivery_time_minutes   R-squared:                       0.595
Model:                               OLS   Adj. R-squared:                  0.595
Method:                    Least Squares   F-statistic:                 1.807e+04
Date:                   Wed, 28 May 2025   Prob (F-statistic):               0.00
Time:                           18:00:28   Log-Likelihood:                 61315.
No. Observations:                 123043   AIC:                        -1.226e+05
Df Residuals:                     123032   BIC:                        -1.225e+05
Df Model:                             10
Covariance Type:               nonrobust
===========================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------
const                       0.0075      0.002      4.406      0.000       0.004       0.011
distance                    0.5711      0.002    250.982      0.000       0.567       0.576
subtotal                    0.2684      0.002    110.778      0.000       0.264       0.273
total_items                 0.0401      0.002     20.736      0.000       0.036       0.044
total_outstanding_orders    0.3854      0.002    184.447      0.000       0.381       0.389
hour                       -0.0692      0.001    -55.549      0.000      -0.072      -0.067
market_id_5.0              -0.0935      0.002    -57.746      0.000      -0.097      -0.090
market_id_2.0              -0.2246      0.001   -171.341      0.000      -0.227      -0.222
market_id_3.0              -0.1046      0.002    -68.859      0.000      -0.108      -0.102
market_id_4.0              -0.1811      0.001   -134.761      0.000      -0.184      -0.178
market_id_6.0              -0.1251      0.007    -17.539      0.000      -0.139      -0.111
==============================================================================
Omnibus:                     1262.895   Durbin-Watson:                   2.001
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1581.485
Skew:                           0.172   Prob(JB):                         0.00
Kurtosis:                       3.436   Cond. No.                         23.9
==============================================================================
```

4.4 Multicollinearity handling : (Reducing features from to take care of VIf) final features

Making sure to take care of multicollinearity and keeping <10 and <5 for most of the features

	Features	VIF
1	subtotal	6.64
0	distance	4.92
2	total_items	4.23
3	total_outstanding_orders	4.00
6	market_id_2.0	2.83
8	market_id_4.0	2.59
4	hour	1.98
7	market_id_3.0	1.40
5	market_id_5.0	1.33
9	market_id_6.0	1.02

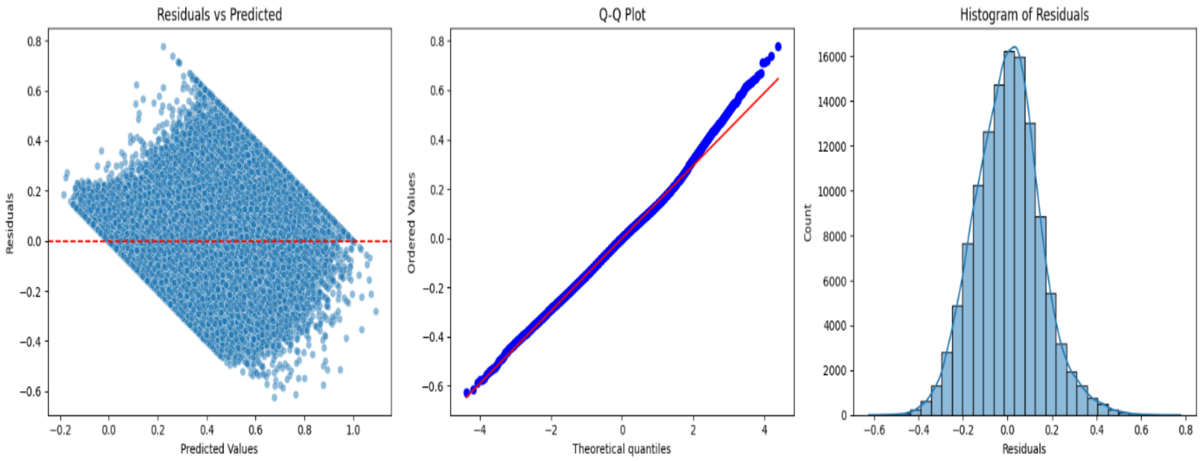
4.5 Coefficient Interpretation on Final Model (Non RFE) picking top important
Each coefficient represents the impact of a unit increase in the corresponding feature on delivery time (since features were scaled, they need to be mapped back to the original scale).

Feature	Coef	Interpretation
const	0.474	When all features are at their minimum (scaled = 0), the baseline delivery time is ~0.0511 (scaled units).
distance	0.3953	Going from min to max distance **increases delivery time by 0.565 units** (in scaled target). Most influential positive factor.

Feature	Coef	Interpretation
subtotal	0.3571	Higher order value (subtotal) correlates with **longer delivery time** (could be due to more complex or bulk orders).
total_outstanding_orders	1.3607	A strong **positive impact**: more unfulfilled orders at the time of request **greatly increases delivery time**.
num_distinct_items	0.1222	More variety in items increases prep time slightly, hence longer delivery.
max_item_price	0.1282	Expensive items might require special prep → **slight increase** in delivery time.
total_busy_dashers	-0.2965	Negative coefficient: more dashers available (i.e., more busy dashers in the area) **decreases delivery time** significantly.
isWeekend	0.0163	Weekend orders take **~0.0163 more time** (scaled) — likely due to higher demand or slower traffic.
created_hour	-0.0031	Orders placed later in the day are associated with **shorter delivery times**, possibly due to lower demand.
market_id_2.0	-0.645	This market has **shorter delivery times** compared to the base category

- Key Takeaways for Support Operations & Optimization
    - Distance is the most significant factor → Implement route optimization to improve efficiency
    - High total outstanding orders increase wait time → Use order prioritization models to prevent backlogs
    - Weekends slightly impact delivery time → Implement dynamic pricing or surge-based incentives for dashers

Insights from the Error Terms Distribution:



Normality of Errors :-

The histogram follows a bell-shaped curve, suggesting that errors are approximately normally distributed.
This is a good sign because linear regression assumes that residuals (errors) are normally distributed around zero.

Mean-Centered Errors

Most of the errors are centered around zero, meaning that the model does not have a significant bias (i.e., it is not consistently underestimating or overestimating delivery time).
There are slight asymmetries on the right side, indicating that some predictions are higher than actual values.

Model Performance Indications

Small error variance
The model has a good fit and predicts delivery times with reasonable accuracy.
Few extreme errors
The presence of longer tails on the right suggests that the model occasionally overestimates delivery time.
If needed, the model can be improved by handling outliers or using non-linear regression models to capture complex patterns.

Recommendations for Improvement:

Analyze High Errors ->
Identify common conditions where the model overestimates delivery times (e.g., peak hours, traffic delays).
Feature Engineering ->
Introduce non-linear features or interactions to capture dependencies.
Consider Other Models->
Explore tree-based models like Gradient Boosting or Random Forest if errors remain significant.

5. Feature impact analysis

5.1. Understanding Coefficients & Their Influence

- Features Increasing Delivery Time (Positive Coefficients):
  - Distance → Most significant factor.
  - Total Outstanding Orders → Delays deliveries
  - Total Items → More items take longer to prepare
  - Weekend Effect → Delivery is slower on weekends
- Features Reducing Delivery Time (Negative Coefficients):
  - Total Onshift Dashers → More available dashers reduce waiting time
  - Total Busy Dashers → High demand accelerates fulfillment

5.2 Regression Equation for Prediction

Delivery Time=0.474+(0.3953×distance)+(0.3571×subtotal)+(1.3607×total_outstanding_orders)+(0.1282×max_item_price)+(-0.0238×total_items)+(−0.8188×total_onshift_dashers)+(−0.2965×total_busy_dashers)+(0.0163×isWeekend)+(0.1222*num_distict_items) $

Recommendations for Operations & Resource Management to Optimize Delivery Time

Based on the analysis of the delivery time prediction model, regression results, error distribution, and feature impact, here are key recommendations to improve operational efficiency, resource management, and delivery time predictions:

- **Improve Delivery Time Estimation & Route Optimization**
  Issue: The model shows high variance, and factors like distance, outstanding orders, and busy dashers significantly impact delivery time.
  Solution:
  - **Implement Dynamic ETA (Estimated Time of Arrival):**

Use real-time data, traffic conditions, and machine learning models to continuously update estimated delivery times.

  - **Optimize Routes Using AI & GPS Data:**

Utilize real-time traffic insights & weather conditions to adjust delivery routes dynamically.

  - **Clustering Deliveries:**

Group nearby orders together to reduce travel time for drivers.

  - **Impact:**

Reduced delivery delays and more accurate predictions, leading to higher customer satisfaction.

- **Efficient Allocation of Delivery Partners**

Issue: The regression analysis shows that total_busy_dashers and onshift_dashers significantly impact delivery time.
When too many dashers are busy, deliveries slow down, indicating a shortage of available delivery partners.

Solution:

- **Dynamic Workforce Allocation:**
  - Use ML models to predict demand surges (e.g., peak hours, weekends) and adjust staffing levels dynamically.
  - Offer real-time shift incentives to onboard more dashers during high-demand periods.

- **Smart Scheduling:**
  - Implement an AI-driven scheduling system that balances shifts based on historical order data & demand forecasts.
  - Reduce idle time by ensuring drivers are optimally assigned based on demand clusters.

Impact: Improved delivery speed and higher driver efficiency while preventing overworking or underutilizing resources.

- **Reduce Delays by Managing Order Volume & Processing Time**

Issue: The feature total_outstanding_orders (1.29) has a high positive coefficient, meaning more unfulfilled orders directly increase delivery time.

Solution:

- **Order Prioritization System:**
  - Prioritize orders based on distance, food preparation time, and delivery partner availability.
  - Implement automated load balancing between different kitchen branches or cloud kitchens.
- **Predictive Inventory Management:**
  - Use ML models to predict order surges and pre-stock high-demand items to reduce kitchen preparation time.
  - Automate order batching and sequencing to reduce processing delays.

Impact: Faster order processing & fulfilment, leading to fewer order backlogs and lower delays.

- **Weekend & Peak-Hour Strategy**

Issue: The "isWeekend" feature (-0.01) has a small but statistically significant impact on delivery time, indicating that deliveries might be slightly slower on weekends.

Solution:

- **Surge-Based Driver Allocation:**
  - Increase delivery personnel availability during peak hours & weekends.
  - Provide incentives or surge pay for drivers working high-demand shifts

- **AI-Driven Demand Prediction:**
    - Use past order trends to anticipate peak hours and adjust resource planning.
    - Enable pre-scheduled orders where customers can book deliveries in advance to reduce last-minute demand spikes.

Impact: Smoother weekend & peak-hour operations, minimizing bottlenecks.

- **Improve Model Performance & Prediction Accuracy Issue:** The error analysis shows a high variance in predictions, meaning the model struggles with accuracy in some cases.

Solution:

- **Enhance Feature Engineering:**
    - Add more contextual features like weather, traffic conditions, and driver experience level to improve prediction accuracy.
- **Switch to an Advanced Model:**
    - Try Gradient Boosting Models (XGBoost, LightGBM) or Neural Networks to improve predictive accuracy.
- **Continuous Model Training:** Deploy an ML Ops framework to retrain models on new data every few weeks to adapt to changing delivery patterns.

Impact: More reliable delivery time predictions, improving customer expectations and reducing complaints.